PROJECT AND TEAM INFORMATION

Project Title

RESQNET: Smart Urban Emergency & Transportation Network Simulator

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

1.Traffic Management Issues :

- There are traffic jams and queues of vehicles at junctions, which cause significant delay of emergency services like ambulances; regular traffic control systems lack priorities to give way to emergency services.
- Traffic is changing all the time, so your travel times and efficiency in choosing routing will be impacted

2.Challenges of Emergency Services :

- Several emergencies may occur simultaneously and in such case it is necessary to choose which of the emergencies require urgent assistance.
- The emergency service, ambulance ,police van has to be used efficiently taking into account their availability, location and the degree of emergency; any delay in choosing appropriate resources and routes will lead to the rise of the emergency response time.

3. The issues connected with navigation and the transport network :

- Not always the quickest route will be the fastest one; there are traffic and congestion which can affect the situation. Current routes may turn out to be unavailable in case of roads closure, accident, and failure of the network.
- Key junctions may influence the connectivity of the entire transport network.

4. Need for integration :

- Emergency response, traffic management and navigation are often considered as separate issues, despite their interdependence; therefore, there is a necessity for an integrated simulation system.

5. Proposed Solution:

- RESQNET aims to overcome these difficulties by concentrating on emergency situations, by providing the necessary resources, by managing traffic and giving emergency vehicles priority, by calculating optimal and alternative routes, by dynamically responding to changes in the road conditions and by analysing the critical points of the transportation network using C++ and Data Structures and Algorithms.

State of the Art / Current solution

- Modern navigation systems work out routes by means of shortest-path and dynamic routing algorithms, considering distance, travel time and changing traffic conditions.
- Smart traffic management systems are now being developed to give emergency vehicles priority by dynamically adjusting the traffic signals and by providing ambulances with faster routes or green corridors.
- When carrying out resource allocation and route planning, advanced emergency medical systems also take into account the availability of ambulances, the level of traffic congestion, hospital capacity and the different kinds of emergency demand. However, even though there are a number of existing methods which concentrate on individual aspects such as traffic control, the routing of ambulances, or the choice of hospital, there is still no single simple system that shows all these interconnected decisions together.
- RESQNET removes this gap by providing a single, integrated simulation model written in C++ and based on DSA which incorporates emergency prioritization, resource allocation, traffic-aware routing, dynamic road updates, hospital selection, and transportation network analysis.

Project Goals and Milestones

#	Milestone	Status
1	Developed rules for the city simulation, including entities and interactions between emergencies, vehicles, intersections, roads, and hospitals	Done
2	Build the base city map and figure out the initial road network with distance and travel-time data	Done
3	Develop emergency incident lifecycle (create, prioritize, dispatch, arrive, close)	In progress
4	Introduce emergency resources and select the most appropriate available resource for an incident	Done
5	Simulate normal vehicle movement and congestion so that road conditions change dynamically during the simulation.	Pending
6	Add handling of traffic at the intersection level and mechanisms to give higher priority of movement to emergency vehicles	Pending
7	Link traffic conditions to route costs so that routing decisions react to simulated congestion, not fixed distances	Pending
8	dynamically reroute when traffic congestion, accidents, or road closures render the current path inefficient or unavailable	Pending
9	Add destination decision logic including selection of a suitable hospital after emergency pick up	Pending
10	Study the city network to find critical junctions and evaluate their failure on emergency access	Pending
11	Run the entire system as a time-based simulation with emergencies, traffic events, movement of resources and failures of the network all occurring simultaneously	Pending
12	Compare various scenarios and assess response time, waiting time, route efficiency, utilization of resources, and network robustness	Pending
13	Complete integrated implementation, testing, documentation, and project demonstration	Pending

Project Approach

- System design: The system will be built as one system that includes Emergency Response Management, Smart Traffic Management and Dynamic Navigation/Transportation Network Analysis.
- The Emergency Response Module will give priority to emergencies and will assign resources based on their availability and how far they are.
- The network modeling will be carried out using a weighted graph, the nodes of which will be intersections, emergencies, hospitals and other important locations and the edges will be the roads together with their travel times and distances.
- The Dynamic Navigation Module will use algorithms based on graph theory to calculate routes for connecting emergency resources with emergency locations and hospitals. It will also update the network whenever there are changes in traffic or when roads are blocked and then calculate the route.
- The Traffic Management Module will simulate traffic at the intersections that have been selected. It will prioritize this by taking into account the number of cars and the time spent.
- A network analysis will be carried out in order to identify the critical junctions that may affect the connectivity of the network.
- The technology and platform employed will be the C++ programming language and the Standard Template Library (STL).
- Implementation will involve the use of graphs, vectors, queues, priority queues, unordered maps, BFS, DFS, and a critical node analysis in the area of Data Structures and Algorithms; as for the development platform, we will at first create a console application by using IDEs such as Visual Studio Code and Code::Blocks.
- The testing method involves, after the implementation has been carried out, testing the system under a variety of conditions, such as emergencies, traffic situations, emergency vehicles, road closures, alternative routes, and network failures.

Project Outcome

• It is a complete console simulator written in C++ and includes modules for Emergency Response Management, Smart Traffic Management and Dynamic Navigation and Transportation Network Analysis.The city is shown using a weighted graph which takes into account roads, junctions, hospitals, emergency points and transport resources.

• The emergency management module is able to generate emergencies and select the appropriate resources from the list of available resources.

• The smart traffic management module models how queues form and how traffic gets backed up. It gives priority to emergency vehicles.

• The dynamic navigation module not only calculates the route but also offers alternative routes if traffic conditions change or roads are blocked.

• The module dedicated to network analysis detects the points or nodes that have an effect on network connectivity.

• The results from the simulation scenarios illustrate the way in which the system performs in emergencies, during traffic jams, in the case of road closures and when there are network disruptions.The last things to be delivered are the source code, the project report or the documentation together with the system design and flow diagrams, the test results and the project presentation.

Assumptions

• The city is viewed as a weighted graph with the nodes standing for places and junctions and the edges representing the roads.

• The traffic situations and emergencies or the road closures are simulated and are not based on real-life data.

• Ambulances and other emergency response vehicles have definite positions, availability levels, and capabilities.

• Some hospitals already have a fixed capacity and facilities.

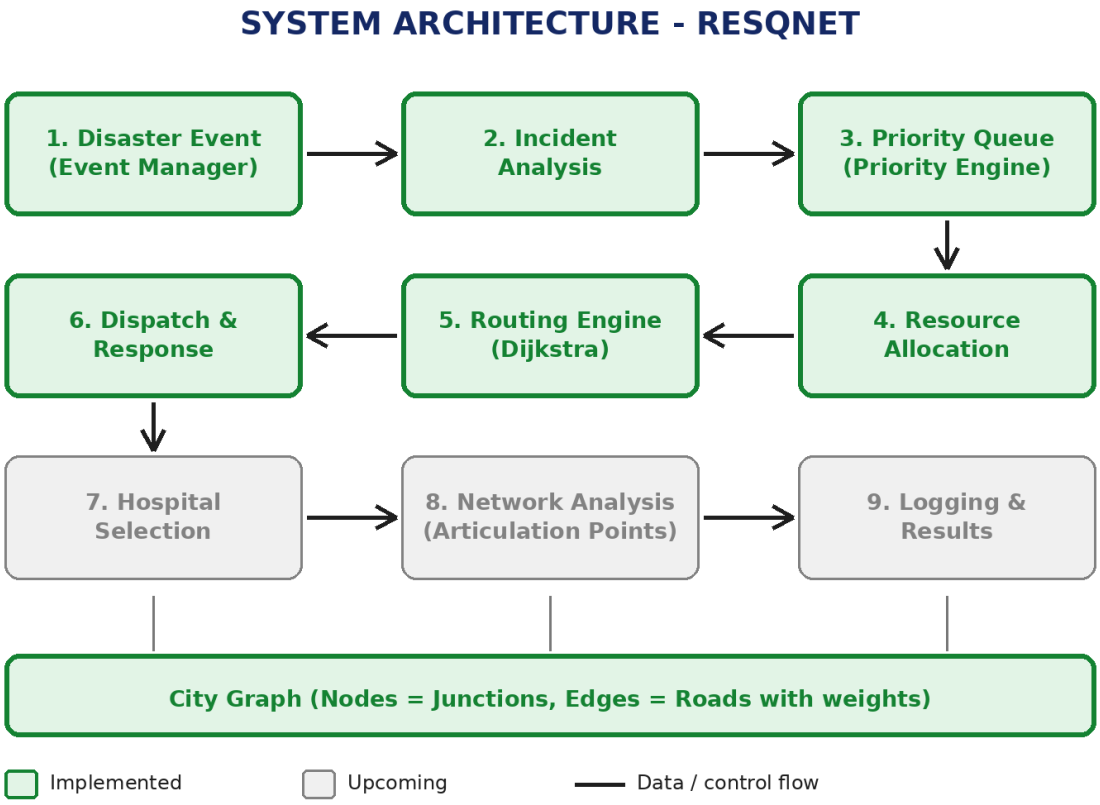
• The degree to which a vehicle is mobile and how long the journey lasts are determined by set values and the present state of traffic.

• Road closures, network breakdowns are simulation scenarios

• The project is about making decisions using algorithms and simulations, not GPS or real-time mapping.

• The project is about algorithmic decision-making and simulation, not involving real-world GPS, live maps, or hardware integration.

System Architecture:



RESOURCES REQUIRED

- Computers that can compile using C++17 along with the necessary building tools needed to run and analyze the simulation. Along with the debuggers, this includes an editor or an IDE like Microsoft Visual Studio Code or Code::Blocks. The C++ STL (Standard Template Library) offers different kinds of data structures that include vectors, queues, priority queues, and unordered maps. Git and GitHub are employed for version control and for allowing team collaboration during the development of the project.
- Include sample data of the city network in order to carry out a realistic simulation, the data covering information on locations, roads, intersections, hospitals, and emergency resources.
- Use synthetic data on emergencies, traffic conditions and road conditions to carry out test scenarios involving multiple emergencies, traffic congestion, road closures and network failures capable of compiling with C++17 (or a more recent version) and the standard build tools needed to implement and test the simulation. An editor or an integrated development environment (IDE) such as Visual Studio Code or Code::Blocks together with the standard debugging tools. The C++ STL (Standard Template Library) includes data structures such as vectors, queues, priority queues, and unordered_maps.

REFERENCES

- E. W. Dijkstra, “A Note on Two Problems in Connection with Graphs,” Numerische Mathematik 1, number 2 (1959).
- C++ Reference Standard Template Library (STL) documentation. Queues. Priority queues. Vectors. Maps (no order). C++ Reference
- GeeksforGeeks: Graph Data Structures and Algorithms, covering Dijkstra's Algorithm, BFS, DFS, Priority Queues and Graph Connectivity
- OpenSSL Project . (Delete this line if your project is not making use of OpenSSL or cryptography.) A version control system and collaboration platform for projects available on GitHub.